

SERIES

```
In [20]: import pandas as pd
a = pd.Series(['divi', 'arthi', 'mouni', 'makizh'])
print(a)
print(a.values)
print(a.index)
print(a.dtype)

#Accessing Elements
print(a[1])

#Access Elements using Loc
print(a.loc[2])
print(a.iloc[2])

#Slicing
print(a[1:3])
print(a.loc[1:3])
print(a.iloc[1:3])
```

```

0    divi
1    arthi
2    mouni
3    makizh
dtype: object
['divi' 'arthi' 'mouni' 'makizh']
RangeIndex(start=0, stop=4, step=1)
object
arthi
mouni
mouni
1    arthi
2    mouni
dtype: object
1    arthi
2    mouni
3    makizh
dtype: object
1    arthi
2    mouni
dtype: object

```

```

In [1]: import pandas as pd
arr = pd.Series([12,25,36,40,50,12],index = ['A','B','C','D','E','F'])
print(arr[['A','C','F']])
print(arr)
print(arr['E'])

#Vectorized Operations - No Loop required
print(arr+5)
print(arr-6)
print(arr*5)
print(arr.loc['A'])
print(arr.iloc[1])

#Statistical Function
print(arr.sum())
print(arr.max())
print(arr.min())
print(arr.std())
print(arr.mean())
print(arr.median())

```

```
print(arr.var())

#Boolean Indexing
print(arr[arr>30])

#Count Values
print(len(arr))
# it return indivitual value count
print(arr.value_counts())

#String Operation
name = pd.Series(['kaviya','jesitha','jenifer'])
print(name.str.upper())
print(name.str.len())
```

```
A    12
C    36
F    12
dtype: int64
A    12
B    25
C    36
D    40
E    50
F    12
dtype: int64
50
A    17
B    30
C    41
D    45
E    55
F    17
dtype: int64
A     6
B    19
C    30
D    34
E    44
F     6
dtype: int64
A    60
B   125
C   180
D   200
E   250
F    60
dtype: int64
12
25
175
50
12
15.523101064757217
29.166666666666668
30.5
240.96666666666664
```

```

C    36
D    40
E    50
dtype: int64
6
12    2
25    1
36    1
40    1
50    1
Name: count, dtype: int64
0    KAVIYA
1    JESITHA
2    JENIFER
dtype: object
0    6
1    7
2    7
dtype: int64

```

```

In [50]: import pandas as pd
a = pd.Series({
    'arthika' : 21 ,
    'riya' : 22,
    'kowsi' : 20 })
print(a)

print(a["riya"])

```

```

arthika    21
riya       22
kowsi      20
dtype: int64
22

```

```

In [66]: #NaN - float type value
#index alignment
s1 = pd.Series([10,20,30],index=['A','B','C'])
s2 = pd.Series([1,2,3],index=['B','C','d'])
print(s1+s2)

s = pd.Series([1,2,3,4,5])

```

```
print(s.count())
print(s.unique())
print(s.nunique())
```

```
A      NaN
B      21.0
C      32.0
d      NaN
dtype: float64
5
[1 2 3 4 5]
5
```

DATA FRAME

In [41]: `import pandas as pd`

#Method 1 : From Dictionary

```
data = ({
    'name': ['arthi', 'mouni', 'divi'],
    'roll_no': [4, 18, 7],
    'age' : [21, 22, 21]})
df = pd.DataFrame(data)
print(df)
```

#Method 2 : From series to DataFrame

```
s1 = pd.Series([1, 2, 3])
s2 = pd.Series([4.9, 5.4, 6.3])
s3 = pd.Series([True, False, True])
df= pd.DataFrame({'sno' : s1,
                  'cgpa' : s2,
                  'attendance' : s3})
print(df)
```

#List of List

```
data = [['dharshini', 22, 98],
        ['dharani', 21, 88],
        ['abi', 20, 77]]
df=pd.DataFrame(data, columns=["name", "age", "mark"])
print(df)
```

#List of Dictionary

```
data =[
    {"name" : "bhuvana","course": "cs","grade":"A"},
    {"name" : "vicky","course": "bca","grade":"A++"},
    {"name" : "ragu","course": "cs","grade":"A"}]
df = pd.DataFrame(data)
print(df)
```

```
data =[
    {"name" : "bhuvana","course": "cs","grade":"A"},
    {"name" : "vicky","course": "bca","grade":"A++"},
    {"name" : "ragu","course": "cs","grade":"A"}]
df = pd.DataFrame(data,index = ["std1","std2","std3"])
print(df)
```

#Adding one extra Columns

```
df['marks']=[99,77,98]
print(df)
```

#Adding one extra Row

```
std4=pd.DataFrame([{"name":"kathir","course": "cs","grade":"A+","marks" : 90}],index=["std4"])
df=pd.concat([df,std4])
print(df)
```

#Create Multiple Rows

```
students = pd.DataFrame([{"name":"saran","course": "ds","grade":"A+","marks" : 90},
                          {"name":"sarath","course": "aids","grade":"A","marks" : 99}],index=["std5","std6"])
df = pd.concat([df,students])
print(df)
```

#Column Access by Bracket Notation and Dot Notation

```
print(df['marks'])
print(df.course)
print(df[['grade','marks']])
```

#shape attribute

```
print(df.shape)
```

#Find Number Of Rows

```
print(df.shape[0])
```

```
#Find Number Of Columns
print(df.shape[1])

#Dtype Attribute
#Used to check Datatype of all columns
print(df.dtypes)

#Info
print(df.info())

print(df.describe())

#Select Specific Rows
print(df.loc["std3"])

#Multiple Rows
print(df.loc[["std1", "std4"]])

print(df.iloc[1])

#Update Value
df.loc['std1', 'grade'] = 'B++'
print(df)
df.loc['std1', 'name'] = 'Surya'
print(df)

#Delete Column
df.drop('grade', axis=1, inplace=True)
print(df)

#Delete Row
df.drop('std6', axis=0, inplace=True)
print(df)

#Sort Values
df.sort_values('marks', inplace=True)
print(df)

#Decending
df.sort_values('marks', ascending = False, inplace=True)
print(df)
```



```
#Statistical Function for particular Column  
print(df['marks'].max())  
print(df['marks'].min())  
print(df['marks'].mean())  
print(df['marks'].median())  
print(df['marks'].std())  
print(df['marks'].mode())  
print(df['marks'].var())
```

	name	roll_no	age
0	arthi	4	21
1	mouni	18	22
2	divi	7	21

	sno	cgpa	attendance
0	1	4.9	True
1	2	5.4	False
2	3	6.3	True

	name	age	mark
0	dharshini	22	98
1	dharani	21	88
2	abi	20	77

	name	course	grade
0	bhuvana	cs	A
1	vicky	bca	A++
2	ragu	cs	A

	name	course	grade
std1	bhuvana	cs	A
std2	vicky	bca	A++
std3	ragu	cs	A

	name	course	grade	marks
std1	bhuvana	cs	A	99
std2	vicky	bca	A++	77
std3	ragu	cs	A	98

	name	course	grade	marks
std1	bhuvana	cs	A	99
std2	vicky	bca	A++	77
std3	ragu	cs	A	98
std4	kathir	cs	A+	90

	name	course	grade	marks
std1	bhuvana	cs	A	99
std2	vicky	bca	A++	77
std3	ragu	cs	A	98
std4	kathir	cs	A+	90
std5	saran	ds	A+	90
std6	sarath	aids	A	99

std1	99
std2	77
std3	98
std4	90
std5	90
std6	99

```

Name: marks, dtype: int64
std1      cs
std2      bca
std3      cs
std4      cs
std5      ds
std6      aids
Name: course, dtype: object
      grade marks
std1      A      99
std2     A++      77
std3      A      98
std4     A+      90
std5     A+      90
std6      A      99
(6, 4)
6
4
name      object
course     object
grade     object
marks      int64
dtype: object
<class 'pandas.core.frame.DataFrame'>
Index: 6 entries, std1 to std6
Data columns (total 4 columns):
 #   Column  Non-Null Count  Dtype
---  -
 0   name    6 non-null        object
 1   course  6 non-null        object
 2   grade   6 non-null        object
 3   marks   6 non-null        int64
dtypes: int64(1), object(3)
memory usage: 240.0+ bytes
None

      marks
count  6.000000
mean   92.166667
std     8.565434
min    77.000000
25%    90.000000
50%    94.000000

```

75% 98.750000

max 99.000000

name ragu

course cs

grade A

marks 98

Name: std3, dtype: object

	name	course	grade	marks
--	------	--------	-------	-------

std1	bhuvana	cs	A	99
------	---------	----	---	----

std4	kathir	cs	A+	90
------	--------	----	----	----

name vicky

course bca

grade A++

marks 77

Name: std2, dtype: object

	name	course	grade	marks
--	------	--------	-------	-------

std1	bhuvana	cs	B++	99
------	---------	----	-----	----

std2	vicky	bca	A++	77
------	-------	-----	-----	----

std3	ragu	cs	A	98
------	------	----	---	----

std4	kathir	cs	A+	90
------	--------	----	----	----

std5	saran	ds	A+	90
------	-------	----	----	----

std6	sarath	aids	A	99
------	--------	------	---	----

	name	course	grade	marks
--	------	--------	-------	-------

std1	Surya	cs	B++	99
------	-------	----	-----	----

std2	vicky	bca	A++	77
------	-------	-----	-----	----

std3	ragu	cs	A	98
------	------	----	---	----

std4	kathir	cs	A+	90
------	--------	----	----	----

std5	saran	ds	A+	90
------	-------	----	----	----

std6	sarath	aids	A	99
------	--------	------	---	----

	name	course	marks
--	------	--------	-------

std1	Surya	cs	99
------	-------	----	----

std2	vicky	bca	77
------	-------	-----	----

std3	ragu	cs	98
------	------	----	----

std4	kathir	cs	90
------	--------	----	----

std5	saran	ds	90
------	-------	----	----

std6	sarath	aids	99
------	--------	------	----

	name	course	marks
--	------	--------	-------

std1	Surya	cs	99
------	-------	----	----

std2	vicky	bca	77
------	-------	-----	----

std3	ragu	cs	98
------	------	----	----

std4	kathir	cs	90
------	--------	----	----

std5	saran	ds	90
------	-------	----	----

	name	course	marks
std2	vicky	bca	77
std4	kathir	cs	90
std5	saran	ds	90
std3	ragu	cs	98
std1	Surya	cs	99

	name	course	marks
std1	Surya	cs	99
std3	ragu	cs	98
std4	kathir	cs	90
std5	saran	ds	90
std2	vicky	bca	77

99

77

90.8

90.0

8.814760348415605

0 90

Name: marks, dtype: int64

77.7

In []: